

Standard Chartered Plc (HK) (2888)

Digital Assets and Stablecoins: A New Frontier for Banks – Key takeaways from panel discussion at JPM TMT Conference

We hosted Mr Geoff Kot, Global Head of CIB Business Platforms and Partnerships, who oversees digital asset strategy at Standard Chartered Bank, on a panel at our Global TMT Conference. The discussion covered a number of topics, including the potential TAM and key growth drivers of stablecoins, the differences between tokenized deposits and stablecoins, the Web3 revolution, the challenges of stablecoins on monetary policy and STAN's strategy in digital assets and stablecoin ecosystems, etc. We believe that development in digital assets and stablecoins represent a paradigm shift, presenting both opportunities and challenges for banks. As one of the first major banks to implement a comprehensive strategy in this space, we believe STAN stands to benefit from first-mover advantage.

Please note that the speaker's remarks do not represent the views of J.P. Morgan research.

Growth potential of stablecoins and the key drivers

- According to a report by Geoffrey Kendrick, Global Head of Digital Asset Research at STAN, stablecoin circulation is projected to reach US\$2trn by 2028 (vs current market cap of stablecoin at ~US\$300bn).
- This growth is set to be driven by the expansion of user cases in cross-border payments, settlement of tokenized Real-World Assets (RWA) and the use of stablecoins as a store of value, etc. Per data from a third-party advisory firm, Mr Kot noted that payments orchestrated by AI agents could reach US\$30 trillion by 2030, which may be transacted via some forms of programmable money.
- Specifically on payment, Mr Kot believes stablecoin offers two advantages over traditional bank transfers: settlement over the weekend, and same day cross-border payment from West to East (against time zone difference). However, he believes stablecoins are unlikely to replace the existing payment systems.
- The speaker highlighted that another growth driver is the acceleration of stablecoin legislation, such as in Hong Kong SAR, UAE and the US. However, progress is uneven across different jurisdictions, with some more restrictive than others; for example, the UK proposes that stablecoin issuers pledge 40% of reserve assets with the Bank of England at zero interest. On the other hand, the US government's more proactive stance on stablecoins and digital assets may support the growth of US-regulated stablecoins.

See page 6 for analyst certification and important disclosures, including non-US analyst disclosures.

J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

Overweight

2888.HK, 2888 HK

Price: HK\$160.30

20 Nov 2025

Price Target: HK\$190.00

PT End Date: 31 Dec 2026

Banks & Financial Services

Katherine Lei ^{AC}

(852) 2800-8552

katherine.lei@jpmorgan.com

J.P. Morgan Securities (Asia Pacific) Limited/
J.P. Morgan Broking (Hong Kong) Limited

Gokul Hariharan ^{AC}

(852) 2800-8564

gokul.hariharan@jpmorgan.com

J.P. Morgan Securities (Asia Pacific) Limited/
J.P. Morgan Broking (Hong Kong) Limited

Lincoln Yu

(852) 2800 8523

lincoln.yu@jpmorgan.com

J.P. Morgan Securities (Asia Pacific) Limited/
J.P. Morgan Broking (Hong Kong) Limited

Haomin Chen

(86-21) 6106 6347

haomin.chen@jpmorgan.com

SAC Registration Number: S1730524080002

J.P. Morgan Securities (China) Company
Limited

Kian Abouhossein

(44-20) 7134-4575

kian.abouhossein@jpmorgan.com

J.P. Morgan Securities plc

Tokenized deposits vs stablecoins

- **Tokenized deposits and stablecoins will coexist:** The speaker highlighted that tokenized deposits operate within bank-controlled ecosystems (intranet), whereas stablecoins function on public chains. In the near to medium term, both are expected to coexist in the ecosystem. STAN is actively pursuing both stablecoin and tokenized deposit initiatives as part of its digital asset strategy.
- **Consideration between tokenized deposits and stablecoins:** Some customers may prefer tokenized deposits due to established banking relationships. But another important factor is the intended use of funds post-transaction. For example, if clients wish to convert programmable money back to fiat currency after transaction, then tokenized deposits are more efficient; if clients prefer to retain programmable money on chains, stablecoins offer greater flexibility. Thus, Mr Kot believes increased on-chain economic activity or scaling of tokenized RWAs would be positive for stablecoin adoption.

Web3 Revolution: Paradigm shift underway but not yet realized

- **Revolutionary moment yet to arrive for tokenized asset trading:** Mr Kot noted that the term “tokenized” is often misapplied, as most existing applications simply mirror the Web 2 ledgers on-chain rather than representing true Web 3 assets. Mr Kot believes the transformative shift will occur when the golden source of truth resides on-chain, enabling instant collateral movement and improved capital efficiency.

Key challenge of stablecoin to monetary sovereignty

- **Key challenge for policymakers and potential mitigation:** Some researchers highlight that smaller, less resilient economies may risk losing monetary policy flexibility or sovereignty as dollar-denominated stablecoins proliferate. Local governments may consider issuing local currency stablecoins to mitigate these risks, and maintaining a balance between local and USD stablecoins is essential for systemic stability. Mr Kot noted that STAN is actively engaging with regulators in its operating jurisdictions to anticipate and address these challenges.
- **The balance of capital control and technology trends:** There seems to be an inherent contradiction in developing stablecoins within regions with strict cross-border capital controls. While technology is borderless, national laws can impose boundaries. Indeed, stablecoins can be used to circumvent capital controls, but this would be illegal and subject to regulatory consequences. Additionally, Mr Kot believes fully on-chain operations remain a distant prospect, with fiat movement in the background still necessary in most cases.

STAN's positioning in digital asset ecosystem

- **A front-runner in digital assets adoption:** STAN is a pioneer among major banks in developing digital assets initiatives, including offering institutional crypto trading in the UK and applying for a stablecoin issuance license via a subsidiary in Hong Kong. Mr Kot noted that the bank is willing to make strategic investments, and its core banking business would collaborate to add value for traditional clients through successful digital asset initiatives. The speaker views blockchain as essential infrastructure for moving from T+1 to T+0 settlement, with additional regulatory benefits in tracing financial crime and quarantining illicit assets. The primary challenge is re-engineering processes and talent to fully leverage the technology, rather than the technology itself.
- **Competitive advantage of STAN vs other ecosystem participants:** The speaker highlighted STAN's reputation, large customer base, and global footprint as key strengths relative to peers in the digital asset ecosystem. Technology companies may emerge as both partners and competitors, driving new value creation across the ecosystem.

JPM views on the panel

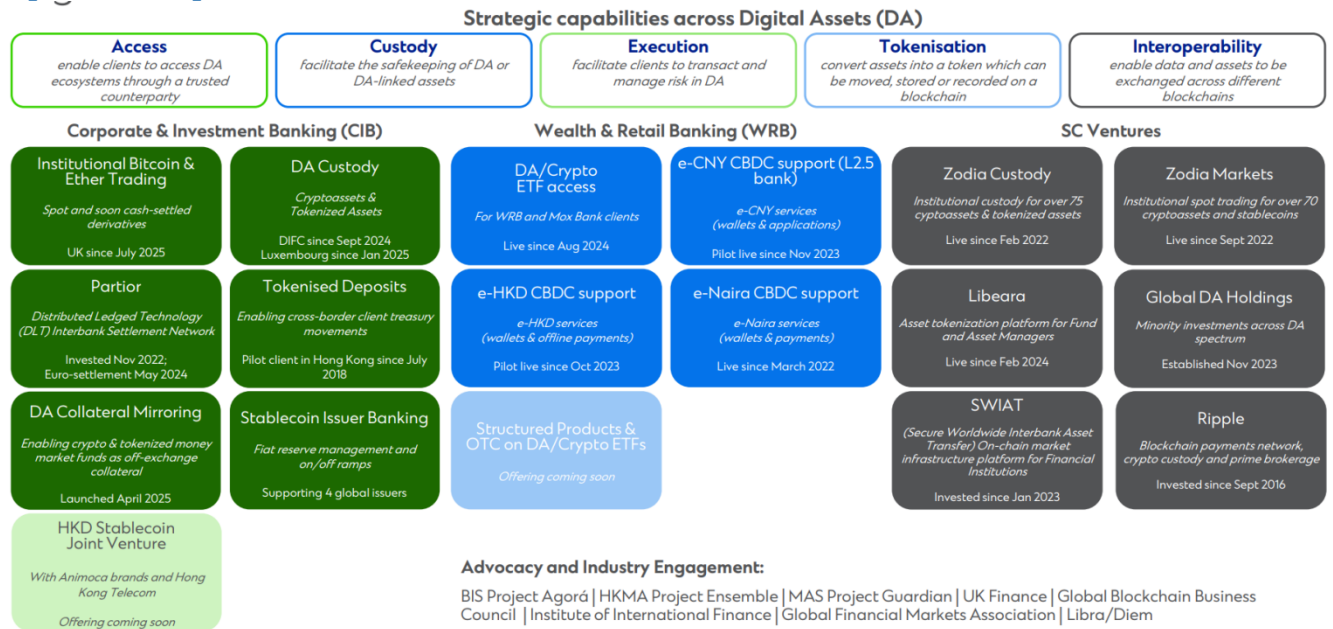
- **Financial perspective – this is not a competition that banks can opt out of:** We expect digital assets (DA) and stablecoin user cases to scale meaningfully this decade, creating both challenges and opportunities for incumbent financial institutions. Banks that take a reactive approach may face greater risks of losing business (clients, market share in payments and deposits) to peers or technology players. Banks do have competitive advantages over other players in the DA or stablecoin ecosystem, including strong networks, a deep understanding of clients' business needs, regulatory compliance capabilities, strong reputation and capital. The key disadvantage for banks is their relative inflexibility in improving user experience and product innovation.
- **STAN set to benefit from first-mover advantage in this new frontier:** One of the key advantages of STAN is its extensive network across over 50 jurisdictions, although one of the key disadvantage is its relatively weak deposits franchise in its five principal markets (HK, SG, UK, Korea, US). Nevertheless, we believe digital assets represent a new frontier, and by leveraging its network advantage and first-mover status among major banks, we believe STAN may improve its market share among existing and new corporate and financial institution clients.
- **Tech perspective – stablecoins/tokenization today don't drive much tech hardware demand, but an increasing agentic AI presence could be a game changer:** Unlike Bitcoin mining (which is a competitive mining process that requires a lot of computing), stablecoin adoption and processing volume increase has not had a big impact on the tech hardware ecosystem. But agentic AI adoption, along with tokenization / stablecoin, could result in much more robust computing infrastructure support. As seen so far in other applications, even a small uptick in agentic AI adoption usually drives significant demand for AI computing.

Table 1: STAN's major involvements in stablecoin and digital asset ecosystem

Entity	Position in the ecosystem	Jurisdiction	Status
JV established by STAN, Animoca, and HKT	Issuer; Participated in HKMA's stablecoin sandbox program and intends to apply for the license for stablecoin issuance	HK	Was sandbox testing
Zodia Custody	Custodian; Provide digital assets custody solutions for institutional clients	UK, Ireland, HK, Singapore	In operation
Zodia Markets	Exchange; An institutional digital asset trading business	UK, Ireland, Jersey, UAE	In operation
Libeara	A tokenization platform helping organizations create, issue, and manage real-world assets on-chain	Singapore	In operation
STAN	Broker; Digital assets trading service which includes spot trading for Bitcoin (XBT/USD) and Ether (XET/USD) for institutional clients	UK (plans to expand to HK and Singapore)	In operation

Source: Company report/announcements.

Figure 1: STAN's digital asset initiatives



Source: Company report.

Investment Thesis

We see StanChart as undervalued relative to its sector/EM peers at 1.1x PTNAV with significant earnings growth potential supported by structural growth drivers and relatively high capital return. We expect a ~7% revenue CAGR over '23-26E, driven by non-NII led by Wealth and Financial Markets, with the group's previous investments producing revenue momentum. NII is likely to be under pressure in '25-26E owing to lower rates, although this would be partly offset by volume growth, increased structural hedging and assertively managed deposits pass-through. We estimate a ~4% cost CAGR with the group's restructuring program likely to drive material cost saves by FY26, supporting a PPOP CAGR of 12% and RoTE reaching 13.8%/13.5% in FY25E/26E. Capital remains strong; we forecast distributions of ~\$10.9bn compared to group targets of >\$8bn in '24-26E and expect a further \$1.5bn of share buybacks to be announced with FY25 results. CRE remains a risk, although we see the group as better positioned compared to peers and pro-actively managing higher-risk exposures lower. Maintain OW.

Valuation

Our Dec-26 PT of HK\$190 is derived from our DDM analysis with NAV discounted back to Dec-26 from FY28E and a fair P/B of 1.0x. Our key assumptions are a normalized ROE of 9.1% and a cost of capital of 8.9%.

Risks to Rating and Price Target

The key downside risks to our rating and price target include:

- **Macroeconomic risks:** Being an EM-focused bank, StanChart is exposed to general macroeconomic risk in EM such as slowdowns, inflationary pressures, depreciation of EM currencies and cross-border and political risks. Potential weakness in China macro and regional sovereign continue to be risks towards revenues and impairments in the near term, as well as geopolitical tensions creating a potential headwind.
- **Provisioning outlook:** We expect provisions to grow to above normalized levels by 2025 given growing pressures in the economic environment. An increase or decrease in provisions as a result of the economic outlook or with regards to specific cases poses a risk to our forecasts.
- **Regulatory uncertainty:** Standard Chartered remains exposed to the changes in the regulatory environment in the UK with capital requirements being set by the PRA in the UK. While StanChart has limited UK exposure, the group is still subject to regulatory changes such as Basel 4 and BoE stress tests.

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

- **Market Maker/ Liquidity Provider:** J.P. Morgan is a market maker and/or liquidity provider in the financial instruments of/related to Standard Chartered Plc (HK) (2888) or related entities.
- **Market Maker/ Liquidity Provider (Hong Kong):** J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited and/or an affiliate is a market maker and/or liquidity provider in the securities of Standard Chartered Plc (HK) (2888) or related entities and/or warrants or options thereon, which are listed or traded on The Stock Exchange of Hong Kong Limited.
- **Manager or Co-manager:** J.P. Morgan acted as manager or co-manager in a public offering of securities or financial instruments (as such term is defined in Directive 2014/65/EU) of/for Standard Chartered Plc (HK) (2888) or related entities within the past 12 months.
- **Client:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients: Standard Chartered Plc (HK) (2888) or related entities.
- **Client/Investment Banking:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as investment banking clients: Standard Chartered Plc (HK) (2888) or related entities.
- **Client/Non-Investment Banking, Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-investment-banking, securities-related: Standard Chartered Plc (HK) (2888) or related entities.
- **Client/Non-Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-securities-related: Standard Chartered Plc (HK) (2888) or related entities.
- **Investment Banking Compensation Received:** J.P. Morgan has received in the past 12 months compensation for investment banking services from Standard Chartered Plc (HK) (2888) or related entities.
- **Potential Investment Banking Compensation:** J.P. Morgan expects to receive, or intends to seek, compensation for investment banking services in the next three months from Standard Chartered Plc (HK) (2888) or related entities.
- **Non-Investment Banking Compensation Received:** J.P. Morgan has received compensation in the past 12 months for products or services other than investment banking from Standard Chartered Plc (HK) (2888) or related entities.
- **Broker:** J.P. Morgan acts as Corporate Broker to Standard Chartered Plc (HK) (2888) or related entities.
- **Debt Position:** J.P. Morgan may hold a position in the debt securities of Standard Chartered Plc (HK) (2888) or related entities, if any.
- J.P. Morgan Securities PLC and/or its affiliates is acting as a Financial Advisor to Standard Chartered Plc in connection with the company’s intention to explore alternatives for the future ownership of its aviation finance business as announced on 11 January 2023.

Company-Specific Disclosures: Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan-covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

J.P. Morgan Equity Research Ratings Distribution, as of October 04, 2025

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	50%	38%	12%
IB clients**	50%	48%	35%
JPMS Equity Research Coverage*	48%	40%	12%
IB clients**	73%	69%	52%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.jpmorganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://www.jpmorganmarkets.com>. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissão de Valores Mobiliários (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC).

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE (“JPM SE”), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions (“EEA professional investors”). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a ‘Research Analyst’ having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpnipl.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Pranav Sata; pranav.d.sata@jpmchase.com; +912261573209. Grievance Officer: Ramprasadh K, jpnipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpnipl.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V. and J.P. Morgan Grupo Financiero are members of the Mexican Stock Exchange and are authorized to act as a broker dealer by the National Banking and Securities Exchange Commission.

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of “accredited investors,” “expert investors” or “institutional investors,” as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities

(Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission, and its registered address is 3rd Floor, 20 North Sathorn Road, Silom, Bangrak, Bangkok 10500.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPMML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have

investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2025. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclosure](https://www.msci.com/disclosure)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2025 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised November 08, 2025.

Copyright 2025 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party. It is permissible to use J.P. Morgan Data for internal business purposes only in an AI system or model that protects the confidentiality of J.P. Morgan Data so as to prevent any and all access to or use of such J.P. Morgan Data by any third-party.